

Phase : Requirement analysis phase

Date	29 september 2026
Team ID	SWTID-2026-9366
Project name	Script - controlled ACL
Maximum marks	4

Here is the requirement analysis based on the provided document:

Business Requirement Overview

The goal of this implementation is to enforce granular, record-level data security on a custom ServiceNow table (u_institution_details) using role-based access control (RBAC) and script-controlled Access Control Lists (ACLs). Access permissions (Read, Create, Write, Delete) are restricted based on specific user roles and field values (e.g., Branch = EEE), while system administrators maintain full access.

1. Functional Requirements

1.1 User & Role Management

- Test User Creation: System must allow creation of a dedicated user (User ID: EEEUser, Name: EEE User, Email: eeeuser@gmail.com).
- Role Configuration: System must support custom roles to manage CRUD permissions:
 - bb1: Grants Read access to EEE branch records.
 - bb2: Grants Create permissions.
 - bb3: Grants Write (edit) permissions.
 - bb4: Grants Delete permissions.

1.2 Data Schema & Storage

- Table: Institution Details (u_institution_details), not extending any parent table.
- Fields:
 - Student Roll Number (Auto Number)
 - Student Name (Reference -> sys_user)
 - Faculty Name (Reference -> sys_user)
 - Branch (Choice: ECE, EEE, CSE)
 - Email (String)
 - Phone Number (String)
 - Description (Multi-line String)

1.3 Access Control Logic

- Read Security:
 - Users with the bb1 role can only view records where Branch is EEE.
 - Users without bb1 cannot view any records.
 - System Administrators retain access to view all records regardless of Branch.
- Create Security: Users with bb2 (and bb1) are granted the ability to create new records (displays the New button).
- Write Security: Users with bb3 (and bb1, bb2) can edit existing EEE records via form or list view.
- Delete Security: Users with bb4 (and bb1, bb2, bb3) are permitted to delete EEE records.

2. Technical & Configuration Architecture

Security Operation	ACL Type	Table Name	Required Role	Condition / Script Logic	Expected Behavior
Read	record	u_institution_details	bb1	Condition: Branch is EEE Script: Checks gs.hasRole('bb1') or admin	Restricts record view in lists/forms strictly to EEE records for bb1 users.
Create	record	u_institution_details	bb2	None	Displays "New" button to users with bb2.
Write	record	u_institution_details	bb3	None	Enables inline and form field editing.
Delete	record	u_institution_details	bb4	None	Enables "Delete" action in record menus/list choices.

3. Acceptance & Verification Criteria

1. Read ACL Verification:
 - Impersonate EEE User (with bb1): Open u_institution_details.list. Only records with Branch = EEE should be visible.
 - Impersonate User without bb1: Open u_institution_details.list. "No records to display" message should appear.
 - Impersonate Admin: All branch records (ECE, EEE, CSE) are visible.
2. Create ACL Verification:
 - Assigning bb2 to EEE User causes the New button to appear on the list view.
3. Write ACL Verification:
 - Assigning bb3 enables field modification (e.g., editing Description field to "ACL is Executed").
4. Delete ACL Verification:
 - Assigning bb4 enables the "Delete" UI action under "Actions on selected rows" for targeted EEE records.